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Forecasting gold price using machine learning methodologies

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ABSTRACT

This study investigates the potential of advanced Machine Learning (ML) methodologies to predict fluctuations in the price of gold. The study employs data from leading global stock indices, the S&P500 VIX volatility index, major commodity futures, and 10-year bond yields from the US, Germany, France, and Japan. Lagged values of these features up to 10 previous days are also used. Four machine learning models are used: Random Forest, Gradient Boosted Regression Trees (GBRT), and Extreme Gradient Boosting (XGBoost), to forecast future gold prices. The study finds that the most influential stocks indices for prediction are one-day lagged data of ASX, S&P500, TA35, IBEX, and AEX, as well as U.S. and Japan bonds yields and delayed data of gas and silver. Furthermore, the study's models identify that one-day lagged VIX score and our VIX dummy variable have a significant impact on gold price, indicating that economic uncertainty affects gold prices. The results suggest that incorporating various financial indicators and moving averages can be a powerful tool for predicting future gold prices. GBRT and XGBoost can be valuable models for making informed decisions about gold investments.

1. Introduction

Gold is known to serve as a hedging commodity against stock crashes and high inflation (see for example Adekoya et al. [1]). Gold is also known to serve as a "safe haven" against stock market crashes and economic uncertainty. Researchers have examined, for example, the role of gold in the Covid-19 pandemic and found that investors expanded investment in gold during the crisis (Akhtaruzzman et al. [2], Wen et al. [3]). Wang and Lee [4] tested whether gold is a" safe haven" for exchange rate risks of five major world currencies and concluded that gold could act as a" safe haven" to hedge dynamic risk for the euro, dollar, and pound in the short run, but not for the renminbi and yen. This important role of gold as a" safe haven" commodity for different financial markets crises stresses the need for a reliable forecasting methodology that would enable portfolio managers, hedgers, and financial institutions to combine gold effectively in their portfolios.

Economic uncertainty usually increases the financial market's fear which is commonly measured by the VIX implied stock market volatility index. Prior research has tried to explain the role of Gold, Bitcoin, and the U.S. VIX as hedging tools against economic uncertainty. Shahzad et al. [5] compared the hedging abilities of those assets against the downside movements of the BRICS countries' stock market indices and concluded that bitcoin and gold are weak hedges. However, their results show that gold appears to a higher and more stable diversification benefits in China while the VIX exhibits higher diversification benefits in, Russia, Brazil, India, and South Africa. Fakhfekh et al. [6] found that both bitcoin and gold have remarkable hedging abilities. Their results show that the VIX is the most effective instrument for hedging stock market indices. Elsayed et al. [7] studied the dynamic connectedness of return and volatility spillovers among cryptocurrency, gold, and uncertainty measures. They found that gold is the net receiver of both the return and the volatility spillovers under all market conditions.

Until recently, researchers have used traditional econometrics to predict gold prices using predetermined hypotheses concerning the correlations between gold and other financial assets. However, these methodologies have been upgraded since Artificial Intelligence (AI) and Machine Learning (ML) capabilities have been developed. Those methodologies have produced a revolution in financial assets price forecasting precision far better than ever been achieved before. Moreover, this new approach to research, have is able to identify nonlinear and complex dependencies that cannot be observed by other methods. In the following research we used the following research ML capabilities to explore interdependencies among multiple financial markets to predict gold prices. Our data contains the world's leading stock indices of all 38 OECD countries, commodities futures, leading 10 years' Bonds yields along with the VIX volatility. The uniqueness of the following research is that it tries to predict gold price volatility using a data set that covers three financial markets: Stocks, Commodities, and Bonds. Prior research

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has concentrated on the link between gold prices and other commodities, others have investigated the correlation between gold price and stocks indices. However, non, have included bond yields predictors that serve as an alternative "safe haven" to gold and such a variety of world's leading stocks indices. Such a data scope enables us to evaluate the relative influence of the OECD countries financial data on gold prices. In addition to the use of the raw data, lagged and Moving Averages (MA) are also being used to identify trends and correlations more easily. These three layers of data (raw, lagged, and MA) provide much more information than the raw data itself that was used in prior research.

This study employs advanced machine learning techniques such as Random Forest, Gradient Boosted Regression Trees (GBRT), and Extreme Gradient Boosting (XGBoost) which have demonstrated efficacy in predicting other financial assets. The efficiency of the Random Forest, GBRT, and XGBoost models in predicting gold price fluctuations can be attributed to their ability to capture non-linear relationships between explanatory variables and the target variable. In addition, the models have the capability to handle large datasets with multiple features and can effectively handle missing or incomplete data. Moreover, the models can be optimized through cross-validation and grid search techniques to select the best hyperparameters, thus improving their accuracy. The results of our study have shown that these models were able to accurately predict gold price fluctuations, with the XGBoost model achieving the highest level of accuracy. Furthermore, in order to identify the most relevant variables among a large set of potential predictors, we implemented stepwise selection to reduce the number of predictors of gold price. Additionally, to explore deep and remote dependencies, we used up to 10th order of lagged data of each variable including gold itself. The combination of these methods allowed us to efficiently identify the most influential factors in predicting gold price fluctuations.

We found that the most influential stocks indices are found to be one-day lagged data of ASX, S&P500, TA35, IBEX, and AEX, U.S., and Japan bonds yields, and delayed data of gas and silver. Moreover, the VIX score has an important impact on gold prices suggesting, that economic uncertainty influences gold prices. Moreover, this study identified that GBRT and XGBoost could be useful models for gold price forecasting enabling decision-makers to make informed decisions about combining gold in their portfolios.

2. Literature review

Precious metals price volatility has been a puzzle that many researchers have tried to solve. Zheng [8] explained that precious metal futures have higher returns when investor sentiment is pessimistic rather than optimistic. Bosch and Pradkhan [9] argued that changes in traders' positions have a destabilizing impact on subsequent conditional volatility in gold, silver, and palladium. Qadan [10] claimed that investors' risk appetite and economic uncertainty are driving precious metal price and their volatility. Rehman et al. [11] have identified the safe-haven nature of precious metals against an oil price drop. Other researchers have explored the link between previous metals to other commodities. Wang and Li [12] offered a predictive model of the prices of three commodities: Oil, Corn, and Gold. They used various scales within a singular spectrum analysis to identify causalities between those future markets. Kang [13] examined spillover effects among six commodity futures markets and found that gold and silver are information transmitters to other commodity futures markets.

Since financial markets are characterized by high complexity and volatility researchers are using Support Vector Machines (SVM) systems for price forecasting and portfolio optimizations. The SVM allows errors within the training data resulting in a significantly reduced error in the test data. Another technique that is used to forecasts the price volatility of financial assets is Long Short-Term Memory (LSTM). This methodology contains three main stages, input, forget, and output which controls the flow of information over time. The forget gate allows the system to decide which data is not relevant to financial forecasting and therefore should be ignored. Akyildirim et al. [14] analyzed the predictive power of the SVM model for cryptocurrencies and argued that at daily or minute levels, the SVM classification algorithms reached about 55–65 % of predictive accuracy. Chen et al. [15] used the LSTM model to predict the direction of Chinese stocks and showed a 13 % forecasting improvement compared to other methods. Borovkova and Tsiamas [16] offered an LSTM model for intraday stock predictions based on technical analysis indicators and found that their model performed better than equally weighted ensembles or benchmark models. Fischer and Krauss [17] used the LSTM system to predict the S&P500 index and reported that this type of forecasting methodology is superior to other ML methods and traditional trading strategies.

Several studies have utilized ML methodologies to try to forecast gold prices. Plakandaras and Ji [18], used an algorithm based on the Support Vector Regression (SVR) for gold forecasting. Their findings show a significant increase in returns generated by the trading strategy. Zhang and Ci [19] used a deep belief network model composed of restricted Boltzmann machines for gold price forecasting. Their results show that the model improved accuracy for gold price forecasting compared to the traditional neural network and linear models. Vidal and Kristjanpoller [20] attempted to forecast gold volatility by combining two deep learning methodologies: LSTM added to convolutional neural networks. They documented a substantial improvement compared to the GARCH and LSTM models. Alameer et al. [21] proposed a model that combines the adaptive neuro-fuzzy inference system and genetic algorithm that can forecast copper prices. Their model improved the forecasting accuracy and decreases the mean square error of traditional models. Livieris et al. [22] proposed a deep learning forecasting model for gold price exploiting the ability of convolutional layers along with LSTM layers and concluded that their model could provide a significant boost to forecasting performance. Jianwei et al. [23] combined Independent Component Analysis (ICA) and Gate Recurrent Neural Network (GRUNN) methods to forecast gold prices and concluded that the fluctuation of the gold price will be interrupted by long-term trends, cyclic recurrent factors, and random events. Moreover, their ICA-GRUNN model provides prediction with high accuracy and outperforms benchmark methods. He et al. [24] proposed gold price forecasting using an integration of LSTM and Convolutional Neural Networks (CNN). Their model that enables to capture of the local patterns and abstracting the spatial features outperformed other conventional forecasting methods.

3. Data and methodologies

Our analysis aims to predict the daily price change of gold price using data of major commodities futures, world leading indices, and U.S, Germany, France, and Japan, 10 years bonds yields. The commodities futures are gold, silver, oil, natural gas, and wheat. These commodities represent four commodities major sectors, energy, precious metal, energy, and food. The world-leading stocks indices included all 38 OECD countries as primary data for global stock volatilities. Additionally, we included the VIX volatility indices to provide a measure of market uncertainty. The VIX was used both as a continuous variable and as a dummy variable in which above 20 represents extreme market volatility and below 20 normal market condition. Our data consists of daily prices of the above assets from February 2011 to February 2023. We then split the data into train and test sets on September 26, 2019. For the predictions, we applied four machine learning (ML) methodologies: Four machine learning models are used: Random Forest, Gradient Boosted Regression Trees (GBRT), and Extreme Gradient Boosting (XGBoost), to forecast future gold prices. To conduct our analysis, we took the first differences of each series including the gold itself, and generated lagged series of the first, second, third, and up to the 10th order, resulting in a dataset with 276 features. Furthermore, we also calculated the Moving Averages (MA) of the time series, which is a known technical analysis enabling us to identify market trends more easily than the raw data

itself. A full list of the selected explanatory variables is presented in Appendix 1.

3.1. Regression Tree (RT)

A regression tree is a type of decision tree used in machine learning for predicting a continuous target variable based on several input features. It uses a tree-like model of decisions and possible consequences, specifically target predictions, to go from observations about an item to conclusions about the item's target value. The internal nodes of the tree represent test conditions on features, each branch represents the outcome of the test, and each leaf node represents a prediction for the target value. Yu et al. [25], for example, used various ML methods and found that classification RT can predict accurately credit rating for different debt grades such as speculative, investment, and default. RT is usually used when the tree has many leaves features that interact in a nonlinear complicated way. Moreover, the tree feature presents a recursive partition consisting of terminal nodes. RT formulation is described in Eq. (1) and the sum of square errors for tree T in Eq. (2).

$$Y = \beta_0 + \beta^T X + \gamma X X^T + \epsilon \tag{1}$$

$$S = \sum_{c \in leaves\ (T)} \sum_{i \in c} (y_i - m_c)^2 \tag{2}$$

where: $m_c = \frac{1}{n_c}\sum_{i \in c} y_i$ is the predictor of leaf c.

3.2. Random Forest (RF)

RF regression is an ensemble machine-learning technique for regression problems. It combines multiple decision trees into a single model, where each tree is trained on a random subset of the data and features. The final prediction is made by averaging the predictions of individual trees. The idea behind this approach is that by combining the predictions of multiple trees, the model reduces overfitting and improves the accuracy of the predictions. The random selection of features in each tree also adds to the diversity of the trees and reduces the correlation between them, further improving the overall performance of the model. Ghosh et al. [26], for example, used LSTM networks and RF procedures to forecast the movement of the S&P500 and concluded that their multi-feature setting provided a daily return of 0.64 %. Derbenstev et al. [27] applied RF and Gradient Boosting Machine (GBM) to forecast major cryptocurrency prices and concluded that using ML methodologies can forecast short-term cryptocurrency prices and can be the basis for automated trading systems. For output the ensemble of trees $\{T_b\}_1^B$, the regression function is the follows:

$$\widehat{f}_{rf}^{B}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(X) \tag{3}$$

If the variables are identically distributed with a positive pairwise correlations, the variance can be expressed as:

$$V = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \tag{4}$$

After B trees are grown $\{T(X;\theta_b)\}_1^B$, The RF regression predictor is defined as:

$$\widehat{f}_{rf}^{B}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x;\theta_b) \tag{5}$$

3.3. Gradient Boosted Regression Trees (GBRT)

Gradient Boosting is an ensemble learning method that combines multiple weak prediction models into a single strong model. In the case of regression, the weak models are typically decision trees, and the combination of multiple decision trees is called a Gradient Boosted Regression Tree (GBRT) model.

At a high level, gradient boosting works by iteratively training a sequence of decision trees on the residuals of the previous trees. In each iteration, the algorithm adds a new tree to the sequence that tries to predict the remaining errors that were not explained by the previous trees. The final prediction is the sum of the predictions of all the trees in the sequence. If we denote $T_i(x)$ to be the prediction function of the i decision tree in the sequence of K trees, then the final prediction function may be written as

$$f(x) = \sum_{i=1}^{K} T_i(x) \tag{6}$$

Each decision tree in the sequence is trained to predict the negative gradient of the loss function with respect to the current model's predictions. The loss function used in regression problems is usually the mean squared error (MSE) which measures the difference between the predicted and actual values. The loss function for the i-th iteration can be written as:

$$L_i = \sum_{i=1}^{K} l\Big(y_i, f_{i-1}(x_i) + T_i(x_i)\Big) \tag{7}$$

where L_i is the loss function for the i-th iteration, y_i is the actual value of the target variable for the i-th data point, $f_{i-1}(x_i)$ is the predicted value of the target variable by the model in the $(i-1)$-th iteration, $T_i(x_i)$ is the prediction of the i-th decision tree for the input x_i, and $l(\bullet)$ is the loss function for a single data point, which is usually MSE.

The algorithm then tries to minimize the loss function L_i by adjusting the parameters of the decision tree, such as the split points and the leaf values. Once the i-th decision tree is trained, its prediction is added to the previous trees' predictions, and the process is repeated for the $(i+1)$-th iteration until a stopping criterion is met. Qian et al. [28] used GBRT to improve a firm's financial distress prediction and concluded that GBRT methodology not only improves prediction accuracy but also makes the results more interpretable.

3.4. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a powerful machine learning algorithm used for regression and classification problems. It is an implementation of gradient boosted decision trees designed to optimize both computational efficiency and predictive accuracy. XGBoost works by creating an ensemble of weak prediction models, in this case, decision trees, and iteratively improving their predictions by minimizing a loss function. It does so by using gradient descent optimization on the loss function. XGBoost regression model can be broken down into the following components:

The objective function minimizes the difference between predicted and actual values. The objective function can be written as:

$$L(\widehat{y}, y) = \sum_i l(\widehat{y}_i, y_i) + \sum_k \Omega(f_k) \tag{10}$$

where $\widehat{y}$ is the predicted value, y is the true value, $l(\bullet)$ is the loss function that measures the difference between predicted and true values, f_k is the k-th decision tree, and $\Omega(f_k)$ is the regularization term that penalizes the complexity of the model.

XGBoost builds an ensemble of decision trees by iteratively improving the predictions of the previous trees. Each subsequent tree is built to correct the errors of the previous tree. This process is called gradient boosting. Specifically, the objective function of the next tree is designed to minimize the residual error of the previous tree. To avoid overfitting, XGBoost adds regularization to the objective function. The regularization term includes two components: the first is the L1 regularization term (Lasso regularization) which adds the absolute value of the weights of the decision tree to the objective function. The second is the L2 regularization term (Ridge regularization), which adds the square

of the weights of the decision tree to the objective function. These regularization terms help to prevent overfitting and improve the generalization ability of the model. XGBoost uses the CART algorithm to construct decision trees. At each node, it evaluates all possible splits and selects the one that leads to the largest reduction in the objective function. It then prunes the tree using a second-order Taylor approximation to the objective function to improve computational efficiency. The final prediction of the XGBoost model is the sum of the predictions of all the decision trees in the ensemble, weighted by the learning rate (a hyperparameter that controls the contribution of each tree). Carmona et al. [29] used XGBoost to predict bank failure in the U.S. banking sector. Using this methodology, they identified certain indicators that are associated with a higher risk of bank failure.

4. Results

4.1. Regression tree

In order to predict fluctuations in gold prices, a regression tree analysis was conducted. This type of analysis is used to model the relationship between a dependent variable and one or more independent variables. The pruned tree in Fig. 1 represents the variables and values that resulted in the best predictive performance based on cross-validation. By pruning the tree, the model is optimized to avoid overfitting, which is a common issue with decision tree models. Compared to linear regression models, the regression tree model can identify complex, non-linear relationships between variables. As a result, it provides a useful tool for forecasting gold fluctuations. The pruned tree presented in Fig. 2 offers valuable insights into the most important variables and their relationships, making it a valuable resource for understanding the factors that affect gold prices.

According to Fig. 1, which illustrates the cross-validation optimization process used to determine the optimal size of the regression tree, it was found that a pruned tree with 17 final leaves would provide the best forecast. The plot in Fig. 1 provides a visual representation of the trade-off between model complexity and predictive accuracy, allowing for the selection of the most appropriate tree size. Ultimately, the decision to use a pruned tree with 17 final leaves will allow for the most accurate and effective forecasting of gold fluctuations.

According to the cross-validation pruning shown in Fig. 2, a pruned tree with 17 final leaves was found to produce the best forecast for gold fluctuations. The decision tree algorithm selected the variables of 3 and 5 days moving averages, one and two days lag gold fluctuations, and five days-lagged SMI as they were the most significant in predicting fluctuations in gold prices. These variables allowed the model to capture both the overall trend of the commodity and the influence of past fluctuations on the current fluctuations. The relative test means squared error (RSE) of 0.6401 and test mean squared error (MSE) of 0.000071772 suggest that the model performed well in predicting the target variable in previously unseen data, indicating good generalization performance and accuracy. In Fig. 3, the scatterplot illustrates the relationship between the actual fluctuations of gold prices and the predicted values from the regression tree model. The plot shows that the predicted values are in agreement with the actual values, as indicated by the red line representing the 45-degree line starting from the origin.

4.2. Random Forest

Random Forest (RF) is an ensemble method that combines multiple decision trees to improve the accuracy and robustness of the model. Fig. 4 illustrates the relationship between the mean squared error (MSE) and the number of trees used in the random forest process. As the number of trees increases, the MSE generally decreases, but there comes a point where adding more trees does not significantly improve the performance. In this case, the plot shows that the MSE reaches a plateau after 30 trees. However, the minimum MSE is attained at 463 trees, hence, this is the number of trees that are ultimately used for the analysis (see Fig. 4).

The RF model was employed to forecast gold fluctuations, and the results show that it had moderate accuracy in predicting future gold price movements. The model generated a test relative mean squared error (RSE) of 0.43635 and a test MSE of 0.00004892, indicating a performance improvement compared to the regression tree model. The RSE value of 0.43635 indicates that the model's predictions were relatively close to the actual values, and the test MSE value of 0.00004892 shows that the model's predictions were accurate. Additionally, it's worth mentioning that the random forest model provides information about the importance of each input variable in predicting gold fluctuations. This information can be used to gain insights into the factors that have the greatest influence on gold prices and can guide investment decisions accordingly. Moreover, the model can be continually updated and refined as new data becomes available, allowing for improved predictions over time. Overall, the random forest model proved to be a useful tool in forecasting gold price fluctuations, and the improved RSE value suggests that this model was better suited for this task compared to the regression tree model.

The relationship between the random forest predictions and the actual gold fluctuations is illustrated in Fig. 5, where a 45-degree line is included to emphasize the strength of the relationship between the two variables. The scatter plot reveals a high correlation between the

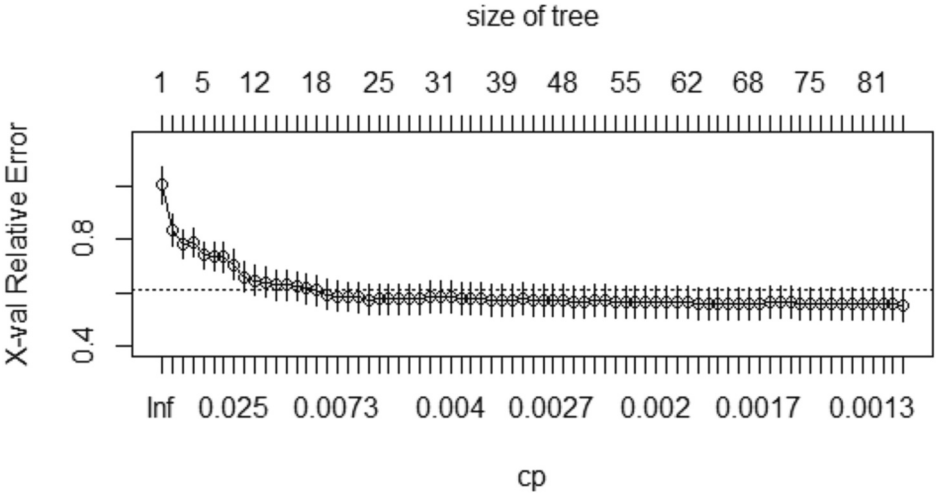


Fig. 1. Optimization to prune the regression tree.

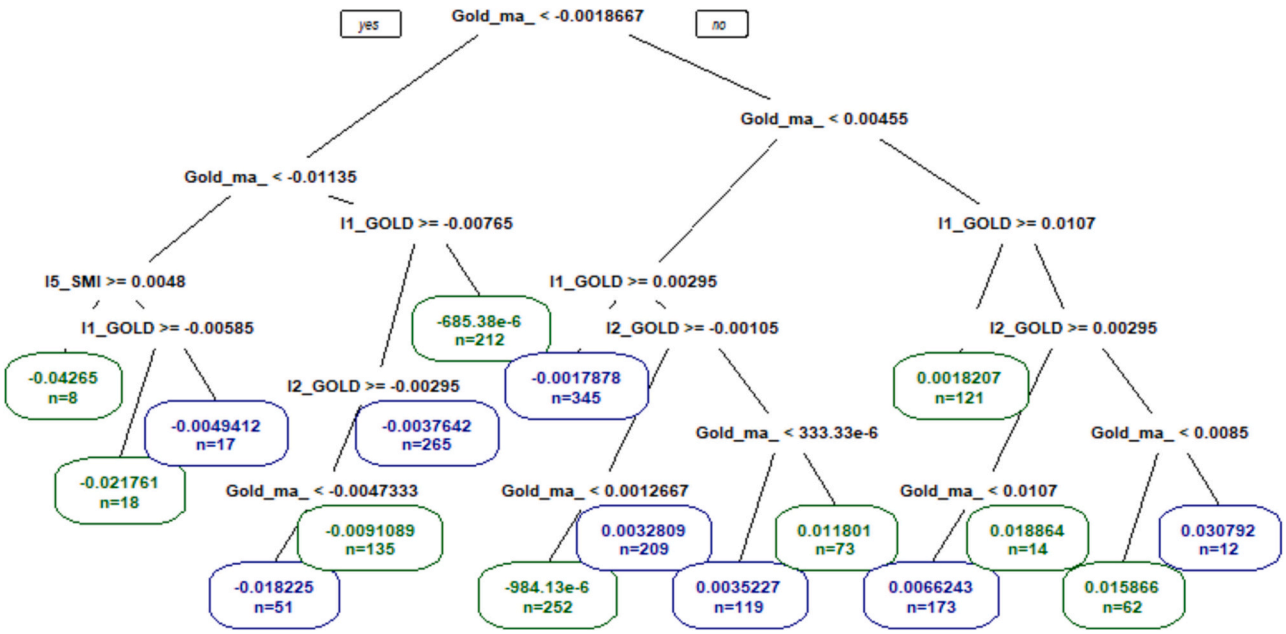


Fig. 2. The pruned regression tree.

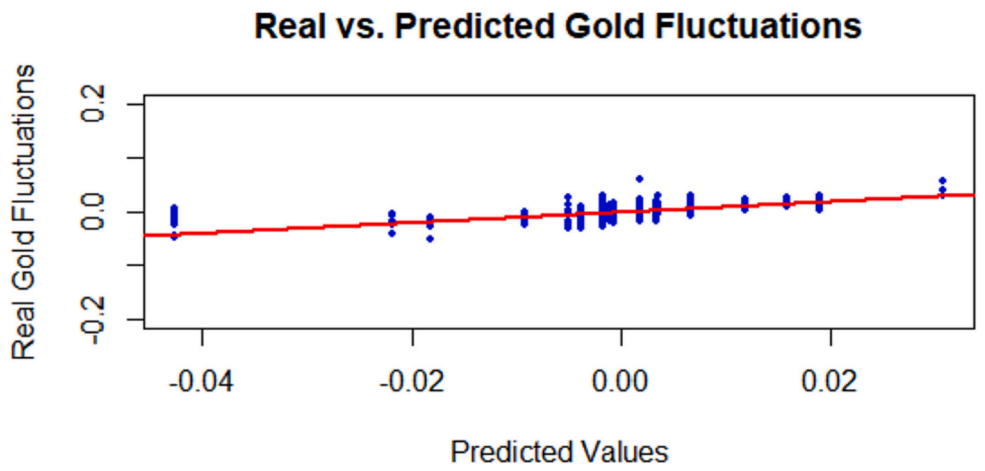


Fig. 3. Scatterplot of real vs. predicted gold fluctuations.

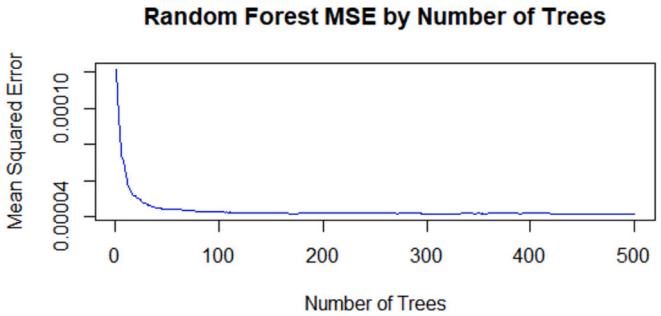


Fig. 4. The optimal number of trees in the RF process.

predicted values from the random forest model and the actual gold fluctuations, with most of the data points lying relatively close to the 45-degree line. This visual evidence supports the model's accuracy in predicting gold price movements and suggests that it can be a valuable tool for making informed decisions about gold investments.

The RF methodology identifies the contribution of each explanatory variable to the accuracy of the random forest model (for the full list see Appendix 2). The model identifies that gold 3 days Moving Average (MA) has the highest contribution to the current daily gold price movement followed by lagged data of gold from 1 to 4 prior days. The most influential stock indices are found to be one-day lagged data of ASX, S&P500, TA35, IBEX, and AEX. Out of the four countries' 10 years bond yields, (US, Germany, France, and Japan) only U.S. and Japan bond yields contributed information to gold price forecasting. The most influential other commodities on gold prices are delayed data of gas and silver (prior 6 and 8 days, respectively). The model has also identified that one-day lagged VIX score and our VIX dummy variable have an important impact on gold price suggesting, that economic uncertainty influences gold prices.

4.3. *Gradient Boosted Regression Tree (GBRT)*

Gradient Boosting is an ensemble learning method that combines multiple decision trees into a single strong model. In this study, we used a five-fold cross-validation technique to optimize for the number of trees and the maximum depth of each tree. Fig. 6 exhibits the outcomes of our

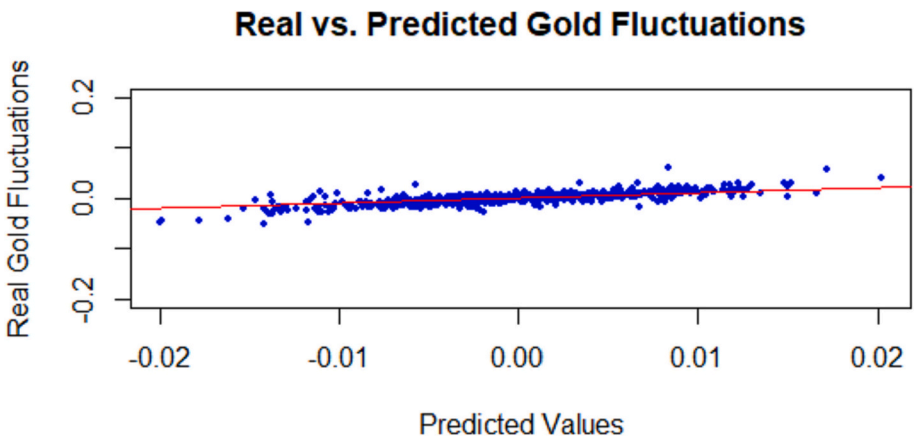


Fig. 5. Scatterplot of real vs. predicted gold fluctuations.

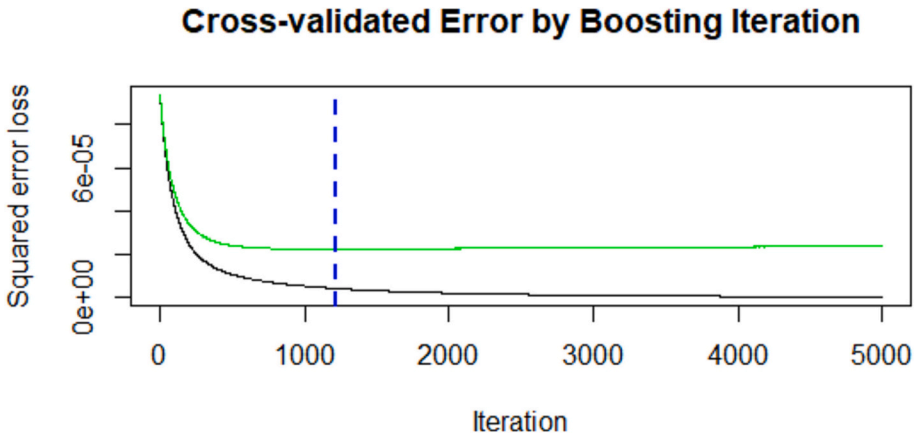


Fig. 6. Optimization for the number of iterations in GBRT.

cross-validation study, indicating that the minimum cross-validation means squared error (MSE) is obtained at 1100 iterations. We selected a maximum tree depth of 15, which was established through grid search as the optimal configuration for our GBRT model. With this optimal configuration, our GBRT model achieved a test MSE of 0.000018605 and a relative MSE of 0.16593, suggesting a significant improvement compared to the random forest model. These values indicate that our model had a high degree of accuracy in predicting future gold price movements.

Fig. 7 displays the relationship between the predicted values from our GBRT model and the actual gold price movements. The plot shows that there is a low variance of distribution around the 45-degree line, indicating that the predicted values closely match the actual values. The scatter plot illustrates that most of the predicted values from our GBRT model fall close to the 45-degree line, suggesting that the model's predictions were highly accurate for a large number of data points.

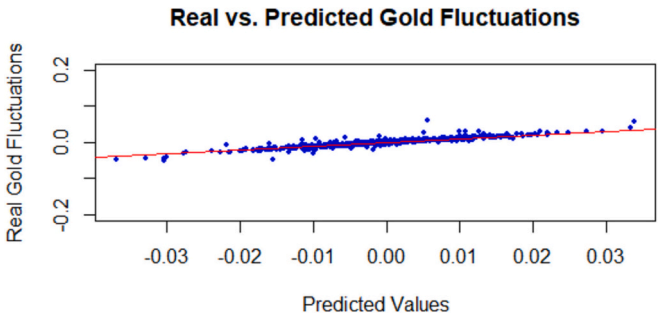


Fig. 7. Scatterplot of real vs. predicted gold fluctuations

4.4. *Extreme Gradient Boost (XGBoost)*

The XGBoost algorithm is a powerful machine learning method that combines the strengths of both gradient boosting and regularization techniques. By incorporating lasso and ridge shrinkage, it can mitigate overfitting and improve generalization performance. In this study, we utilized cross-validation to identify the optimal values for the L1 and L2 regularization parameters. We also used grid search to determine the optimal combination of hyperparameters, including the learning rate and the maximum tree depth. This approach allowed us to obtain the best possible predictive model for gold price fluctuations. The optimal XGBoost model achieved a test mean squared error (MSE) of 0.0000186 and a relative test MSE of 0.1928, indicating its high accuracy in predicting future gold price movements. These results demonstrate the effectiveness of the XGBoost algorithm in capturing complex relationships between variables and the target variable.

Overall, the XGBoost model provides a valuable tool for making informed decisions about gold investments. It's high accuracy and ability to handle complex relationships between variables make it a powerful tool for predicting future gold price fluctuations.

Fig. 8 represents the relationship between the predicted values generated by our XGBoost model and the actual gold price movements. The scatter plot depicts the predicted values on the x-axis and the actual values on the y-axis. The 45-degree line, which represents a perfect match between the predicted and actual values, is also shown on the plot

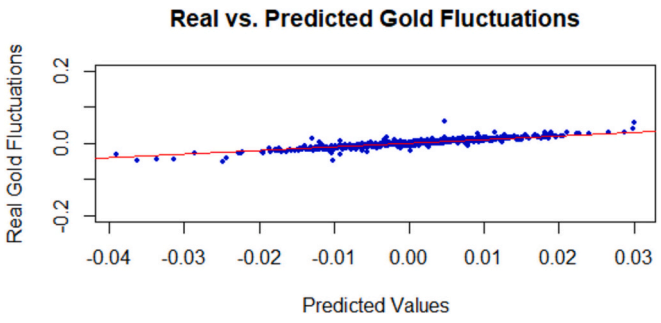


Fig. 8. Scatterplot of real vs. predicted gold fluctuations.

to highlight the strength of the relationship between the two. The plot shows that the predicted values are highly correlated with the actual values, with most of the points lying close to the 45-degree line. This indicates that the XGBoost model's predictions were accurate for a large number of data points. Moreover, the low variance of the distribution around the 45-degree line indicates that the model's predictions were consistently close to the actual values, making it a reliable tool for predicting future gold price movements.

5. Summary and conclusions

In this study, we utilized various ML algorithms, including regression trees, random forests, GBRT, and XGBoost, to predict future gold price movements. The models were trained and tested on the same dataset, and their performance was evaluated using relative test mean squared error (RSE) and test mean squared error (MSE). We used a combination of major stock indices, leading 10 years bonds, and commodities such as oil and wheat as predictors in our models. Our results show that these ML algorithms can be effective tools for predicting gold price movements, with varying levels of accuracy. The examined models identified that gold three days Moving Average (MA) has the highest contribution to the current daily gold price movement followed by lagged data of gold from 1 to 4 prior days. The most influential stock indices are found to be one-day lagged data of ASX, S&P500, TA35, IBEX, and AEX. Out of the four countries 10 years' bond yields (US, Germany, France, and Japan), only U.S. and Japanese bond yields contributed information to gold price forecasting. Gas and silver have been found to have a delayed influence on current gold prices.

Gold along with other financial assets such as the Bitcoin and VIX stock's implied volatility index are known to serve as hedging tools against economic uncertainty. As in prior research, we also find that gold has a crucial diversification ability against market fear and economic uncertainty. We documented that one-day lagged VIX score and our VIX dummy variable have an important impact on gold price suggesting, that economic uncertainty influences gold price.

A comparative analysis of the above results indicates that all models achieved a considerable improvement in accuracy compared to the baseline regression tree model. The random forest model outperformed the regression tree model, achieving a substantial improvement in accuracy. The GBRT model achieved a major improvement in accuracy compared to the random forest, and the XGBoost model demonstrated the highest level of accuracy in predicting gold price movements. Despite the differences in the models' performance, there is a consensus that all models have the capability to accurately predict gold price fluctuations. The high accuracy of the models can be attributed to their ability to capture non-linear relationships between explanatory variables and the target variable and handle large datasets with multiple features.

Our study demonstrates the effectiveness of ML algorithms in predicting future gold price movements. Our models, particularly the GBRT and XGBoost models, achieved high levels of accuracy in predicting gold price movements, providing investors and financial analysts with a more accurate and reliable tool for making informed decisions about gold investments. The combination of raw data along with lagged MA and the VIX volatility index enabled to capture of complex relationships between the three examined financial markets: stocks, commodities, and bonds, providing important insight into the important role of Gold as a hedging asset against economic risks.

The main limitations of our research are the lack of data from developing countries, the variety of bonds that were used, and the absence of cryptocurrencies, Future research should take into account a greater variety of bonds and stocks indices and major cryptocurrencies to further contribute to the knowledge of the role of Gold in the global financial economy.

CRediT authorship contribution statement

The contribution to this publication is equal to both authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix 1

A complete list of the selected explanatory variables from the stepwise selection includes the following: one, two, three, four, five, six, seven, eight and nine days lagged gold fluctuations, three, five and ten days moving averages of gold fluctuations, one, two and seven days lagged S&P fluctuations, one and two days lagged TA35 fluctuations, one and nine days lagged IBEX fluctuations, one, four, five, six, seven, eight and ten days lagged AEX fluctuations, one, four, five, six, seven, eight and ten days lagged fluctuations in BEL, one and eight days lagged ASX fluctuations, one day lagged VIX fluctuations, one, three and 10 days VIX dummy variable that receives the value of one if VIX is greater than 20, one and three days lagged fluctuations in US bonds yields, one day lagged fluctuations in Japan bonds yield, two and ten days lagged fluctuations in NSDQ, two, four, six, seven and eight days lagged Nifty fluctuations, two, three, four, five, six, seven, eight and nine lagged N225 fluctuations, three days lagged BVSP fluctuations, four and eight days lagged MOEX fluctuations, four and five days lagged SMI fluctuations, four and five days lagged HIS fluctuations, six and nine days lagged GAZ fluctuations, six, seven and nine days lagged FTSE fluctuations, seven days lagged SET fluctuations, eight, nine and ten days lagged Silver fluctuations, nine days lagged Wheat fluctuations, ten days lagged DAX and NZX fluctuations.

Appendix 2

The contribution of each explanatory variable to the accuracy of the random forest model is presented in the following table. Please note that the variables with the prefix 'l#_' represents the number of lags taken on that particular variable. The "vix_dummy" is a dummy variable that receives "1" if

the VIX index is greater than 20.

Variable	Contribution
Gold_MA_3	0.049966399
l1_GOLD	0.019207901
l2_GOLD	0.017918855
l1_ASX	0.002232515
l3_GOLD	0.002219208
l4_GOLD	0.001963917
l1_SP500	0.001531251
l1_TA35	0.001477213
l6_GAZ	0.001461528
l3_Usyield	0.001438446
l8_Silver	0.001395272
l1_IBEX	0.001273558
l5_N225	0.00124624
l1_AEX	0.00124352
l5_SMI	0.001188796
l7_N225	0.001165449
l6_Nifty	0.001159611
l7_SET	0.001112221
l9_FTSE	0.001100364
l9_Wheat	0.001091373
l7_Nifty	0.001089446
l7_BEL	0.001087911
l1_Usyield	0.001080972
l9_IBEX	0.001071255
l10_NZX	0.001054144
l2_Nifty	0.001053071
l2_N225	0.001050322
l6_BEL	0.00104733
l7_FTSE	0.001023262
l4_N225	0.001008689
l1_JAPANYield	0.001003315
l4_MOEX	0.001003264
l3_N225	0.000999807
l8_MOEX	0.000996453
l1_VIX	0.000984781
l5_BEL	0.000972938
l2_TA35	0.000961839
l10_DAX	0.000957736
l4_HIS	0.000941529
l4_BEL	0.00093914
l6_FTSE	0.000915598
l5_HIS	0.000900247
l3_BVSP	0.000889518
l2_NSDQ	0.000747899
VIX Dummy	0.0000542713

Note: MA = Moving Average.

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